

IB MATH AA SL

Lesson 12

Composite & Inverse Functions

Functions · 60-minute lesson

Syllabus SL 2.2 · Property of Lynda Badmus Education

f

Do Now — recall

ON YOUR WHITEBOARD

1. $f(x) = x + 3$, $g(x) = 2x$. Find $f(g(1))$.
2. If $f(x) = x + 5$, what undoes it?
3. Make x the subject of $y = 2x - 1$.

Answers (click to reveal)

1. $g(1)=2$, $f(2)=5$
2. subtract 5
3. $x = (y + 1)/2$

What we're learning today

LEARNING OBJECTIVES

- 1 Form**
composite functions $f(g(x))$.
- 2 Evaluate**
a composite at a value.
- 3 Find**
the inverse function $f^{-1}(x)$.
- 4 Understand**
inverse as a reflection in $y = x$.

SUCCESS CRITERIA — I CAN...

- ✓
work a composite from the inside out
- ✓
evaluate $(f \circ g)(x)$ at a number
- ✓
find an inverse by swapping x and y and rearranging
- ✓
state the link between a function and its inverse

Composite functions

INSIDE OUT

$$(f \circ g)(x) = f(g(x))$$

Order matters

do g first, then f

Substitute

put g(x) wherever f has an x

WATCH OUT

$(f \circ g)$ is usually NOT $(g \circ f)$.

Work the inner function first.

Check

brackets keep the substitution tidy.

Form a composite

QUESTION

$f(x) = 2x + 3$ and $g(x) = x^2$.

Find $(f \circ g)(x)$.

SOLUTION

Step 1 $(f \circ g)(x) = f(g(x))$ — put g into f :

$$f(x^2) = 2(x^2) + 3$$

Step 2 simplify:

$$(f \circ g)(x) = 2x^2 + 3$$

Tip:

Replace the x in the OUTER function with the whole inner function.

Your turn — composites

ANSWER IN YOUR BOOK

1. $f(x)=x+1$, $g(x)=3x$. Find $(f \circ g)(x)$.
2. Using the above, find $(g \circ f)(x)$.
3. $f(x)=x^2$, $g(x)=x-2$. Find $(f \circ g)(4)$.

Answers (click to reveal)

1. $3x+1$
2. $3(x+1)=3x+3$
3. $g(4)=2$, $f(2)=4$

Inverse functions

REVERSING f

$$y = f(x) \Rightarrow \text{swap } x, y \Rightarrow f^{-1}(y)$$

Step 1

write $y = f(x)$

Step 2

swap x and y

Step 3

make y the subject — that's $f^{-1}(y)$

PROPERTIES

f^{-1} undoes f : $f^{-1}(f(x)) = x$

Graph

reflection of f in the line $y = x$

Domain $\leftrightarrow$ Range

they swap over.

Find an inverse

QUESTION

Find the inverse of

$$f(x) = 2x - 3.$$

SOLUTION

Step 1 write $y = 2x - 3$, then swap x and y :

$$x = 2y - 3$$

Step 2 make y the subject:

$$y = \frac{x + 3}{2}$$

Step 3 state the inverse:

$$f^{-1}(x) = \frac{x + 3}{2}$$

Exam-style question

$f(x) = 3x + 1$ and $g(x) = x - 4$.

(a) Find $(f \circ g)(x)$. [2]

(b) Find $f^{-1}(x)$. [3]

MARK SCHEME — reveal after students attempt (tutor only)

(a) $f(x-4) = 3(x-4)+1 = 3x - 11$ (M1)(A1)

(b) $y = 3x+1$; swap $\rightarrow x = 3y+1$ (M1); $y = (x-1)/3$ (A1) $\rightarrow f^{-1}(x) = (x-1)/3$ (A1)

Common mistakes



Order of composition

$(f \circ g)$ means g first — not f first.



Inverse $\neq$ reciprocal

$f^{-1}(x)$ is the inverse, not $1/f(x)$.



Swap then rearrange

Don't forget to swap x and y before solving.



Domain/range swap

The inverse's domain is the original's range.

Mixed practice

QUESTIONS

1. $f(x)=x-5$, $g(x)=2x$. Find $(f \circ g)(x)$.
2. Find f^{-1} for $f(x)=4x$.
3. Find f^{-1} for $f(x)=x/2 + 1$.
4. $f(x)=x^2$, $g(x)=x+1$. Find $(f \circ g)(2)$.

Answers (click to reveal)

1. $2x-5$
2. $x/4$
3. $2(x-1)$
4. $g(2)=3$, $f(3)=9$

Plenary & exit ticket

KEY FORMULAS

$$(f \circ g)(x) = f(g(x))$$

f^{-1} : swap x, y then solve

EXIT TICKET — before you leave

1. $f(x)=x+2$, $g(x)=3x$. Find $(f \circ g)(x)$.

2. Find f^{-1} for $f(x)=5x$.

3. What undoes 'add 7 then double'?

Answers (click to reveal)

1. $3x+2$

2. $x/5$

3. halve, then subtract 7

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Composite & inverse functions.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 13

Transformations of graphs — translations, stretches and reflections.

$$y = a f(x - h) + k$$